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A Cost Effective Electronic Braille for Visually Impaired Individuals

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Abstract — Learning process is very difficult for Visually Impaired people as they are unable of receiving visual information. Braille is basically a writing system to aid the visually impaired people. Unfortunately, people of developing or underdeveloped countries cannot afford the available advance electronic braille as they are very expensive. This paper proposes a prototype of affordable Braille display for the blind people to read when an input is given through a computer. Here electromagnetic solenoids are used to facilitate the vertical movement of the dots of the Braille display controlled by an Arduino Uno. This device will be able to guide blind people to read certain patterns of dots in the Braille Cell known as braille alphabets. The study approaches towards the development of a Braille under \$100 which will be portable as well as affordable.

Keywords: *Braille, cell, portable, blind-guiding, partial help, visibility.*

I. INTRODUCTION

Our eyes and brain work together to see; but when this doesn't work the way it should, a person may be visually impaired, or blind. Out of seven billion global populations at present, about 285 million people are visually impaired. Among them 39 million are blind, out of which 19 million are children (aged between one and fifteen years). About 90 per cent of the world's visually impaired people live in developing countries [1].

In developing countries, lack of Braille based educational resources and technological solutions have made the learning process lengthy and complicated for visually impaired students [2]. Available electronic Braille systems are very expensive, ranging \$500- \$2000. Most of the visually impaired or blind people of developing countries like Bangladesh cannot afford Braille due to the costly Braille related technological tools [2].

In fact, the most significant drawback to refreshable braille devices has always been their cost. From the mid-1980s to the present, the loose rule of estimation applied to these products has translated into something like \$100 to \$150 per braille cell. A 40-cell display, in other words, may cost \$4,000 to

\$6,000, while an 80-cell model will cost \$8,000 to \$12,000 [3].

It has been observed, while desirable, braille computer access has been out of reach for many users of assistive technology. The objective of this paper is to design an electric Braille for the welfare of visually impaired people at a very low cost which will help them to read with less difficulties.

This paper aims to develop a prototype portable Braille which can assist visually impaired people in reading. The portable braille cell is developed by using 6 electromagnetic solenoids, 6 MOSFETS and 1 Arduino UNO. This device is powered by a 5 Volt battery.

II. METHODOLOGY

A. Technique:

In this section working of a single cell of the proposed Braille display is explained. An input is given from a computer which is processed by the Arduino Uno. The Arduino Uno controls the switching circuit which is connected to the solenoids. As per the electrical signals from Arduino Uno the solenoids move independently forming different patterns to form the Braille Alphabets. Then these patterns are sensed by finger and a visually impaired or blind user can read the inputs through a computer.

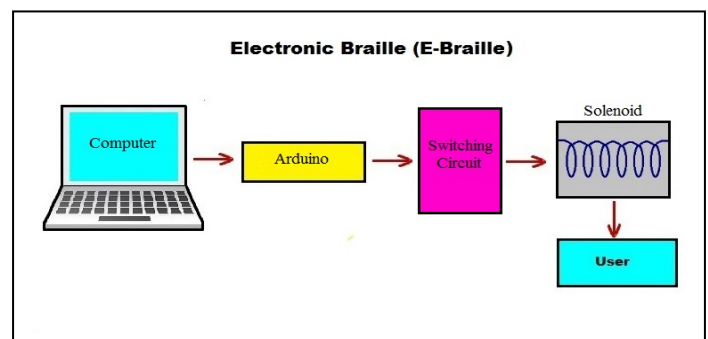


Fig. 1. Block diagram of proposed E-Braille

B. Overview of Mechanical Design

Microcontroller – Arduino UNO was chosen as the microcontroller board because it can be easily programmed. It

needs 5 Volt to operate. For this E-Braille system, Arduino receives character or number inputs from a computer.

MOSFET - P-Channel Logic MOSFET NDP6030PL was used which works as a switch when Arduino supply gate voltage.

Solenoid – Electromagnetic solenoids move to create the corresponding formations of the Braille.

In order to create an electromagnetic solenoid, copper wires were wound on plastic sticks. The copper wires consist of 38 gauges. After winding the copper wire, an iron pin was inserted into the electromagnetic solenoid which mainly works to facilitate the vertical movement of the dots of the Braille cell.

The 6 Electromagnetic solenoids were attached together in 2 parallel columns with 3 solenoids in each column to form a Braille cell. The one end of each solenoid are connected to the positive terminal of the 5volt battery and the other end of each solenoid are connected to the drain of individual MOSFETS. The gate of the each MOSFETS were connected to different output digital pins of the Arduino UNO and the source of the MOSFETS were connected to the negative terminal of the battery. The p-type MOSFETS work as a switch. When an input is given from the computer to the Arduino, it converts the input and gives an electric signal to the output digital pins. This electric signal switch on the MOSFET which eventually allows electricity to pass through the solenoids causing them to move vertically upward and change the position of the dots forming the desired patterns which is then read by the user.

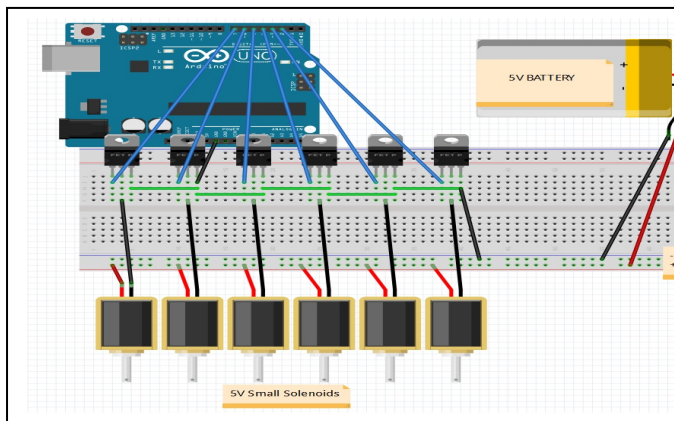


Fig. 2. E- Braille incorporating with Arduino, MOSFET and Solenoid . The Cicut was drawn in Fritzing Software.

III. COST ANALYSIS

National Federation of Blind (NFB United States) have been reported several list of comprehensive index of software systems, devices and technology updates [4]. Though varieties of tools are available for visually impaired people but they are produced in a limited quantity which inherently leads to high costs due to a small market base [5]. The prices of available electronic Braille are \$500-\$2000 according to their features

[6]. The cost of refreshable Brailles has been always a concern as majority of the people cannot afford them due to high price.

Cost Table		
Component	Quantity	Price (\$)
Arduino Uno	1	7.5
MOSFET	6	4.2
EM Solenoids	6	29.75
Others(Battery,Wires,etc)		\$2
Total		43.45

Fig. 3. Table of Cost. All components are locally purchased

The aim of this study is to design a cost effective Braille which can be affordable by the people of developed or underdeveloped country where 90% of the visually impaired people are living.

At present, the lowest cost is \$62 per cell [3]. The study has helped us to drastically reduce the price to \$43.45 which is approximately 29.93% lower per cell. Approximately 68.5% of the total cost of the cell is due to the electromagnetic solenoid. This cost will drastically fall when the solenoids are made locally. Moreover in case of mass production of this braille cell the cost will be even lower.

IV. RESULTS & DEMONSTRATION

a	n	#	=	\$
b	o	1	&	%
c	p	2	!	?
d	q	3	)	+
e	r	4	(	-
f	s	5	]	@
g	t	6	[	^
h	u	7	<	_
i	v	8	>	"
j	w	9	:	.
k	x	0	\	;
l	y	'	/	space
m	z	,	*	

Fig. 4. Braille characters. Inputs were given a computer platfor, the solenoids moves vertically according to the input. For Example when we give "a" as input, 1st solenoid of 1st column moves vertically up [7].

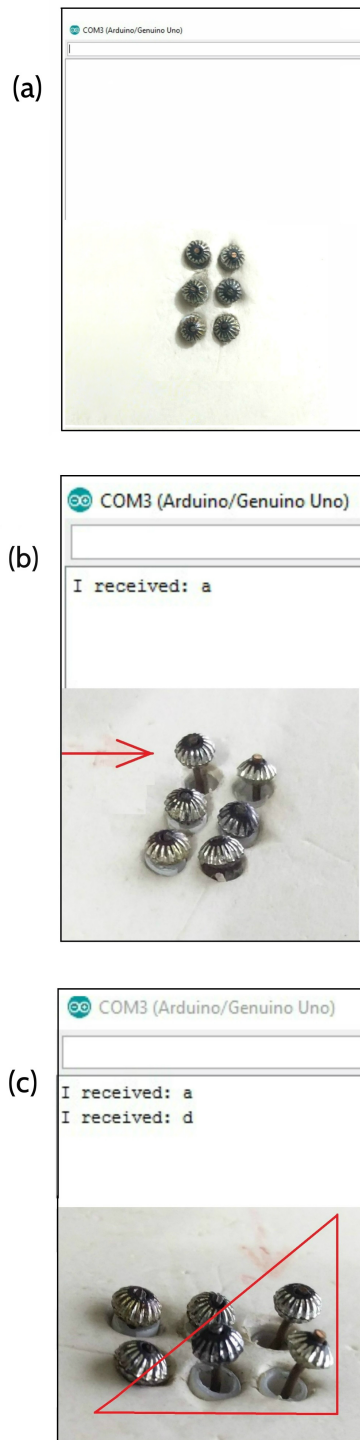


Fig. 5. (a) when no input is given (b) When “a” is given as input in Arduino IDE, the 1st solenoid of 1st column moves vertically up. A red arrow is used to show the movement. (c) When “d” is given as input 3 solenoids moves vertically up. A red triangle is used to show the position change of these 3 solenoids.

The electromagnetic solenoids work perfectly to show the desired formations according to the corresponding character input. The device successfully converted alphabets, numbers and symbols into Braille codes.

Braille symbols are formed within units known as Braille cells. A full Braille cell consist of six raised dots arranged in two parallel columns each having three dots. The dot positions are identified by numbers from one through six. Whenever an input is given on a computer platform (Arduino IDE) the solenoids moves vertically up and the dot positions are changed. The user can easily detect the alphabet, numbers or symbols by touching the cell with their figure tip. The Braille can be reset by using the “space” key of the keyboard.

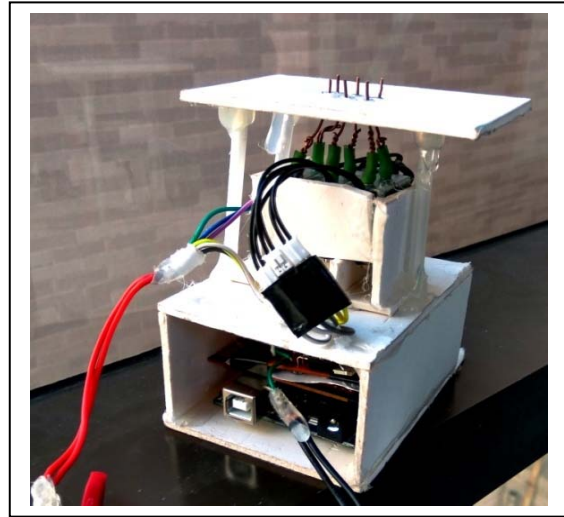


Fig. 6. E-Braille designed with Arduino, MOSFET and Solenoid.

An experiment was conducted on 12 subjects who can read Braille alphabets. A set of few random alphabets and numbers were displayed on the braille cell for the subjects to read and the results of successful readings were recorded. The results of the experiment are shown below.

Alphabets Numbers	Number of people who read successfully (out of 12)
a	12
d	12
l	12
q	11
w	12
y	10
0	11
3	12
7	12

Fig. 7. Table showing experiment results.

The results of the experiment were quite satisfying. Most of the participants were able to read the characters successfully from the dots interface of the Braille.

V. DISCUSSION

A. Challenge

Finding the appropriate power supply for the Braille cell was a challenge. As the actuation of solenoids depends on the current mostly, two batteries of the appropriate (5V) voltage had to be connected in parallel direction; as batteries of higher voltage would severely damage the solenoids.

Another issue was the heating problem of Solenoids. The solenoids got heated if current was conducted through them for a long time; this could potentially cause permanent damage them permanently. To avoid any damage a time of 20 seconds was set after which the current supply in the solenoids would automatically be stopped.

During the experiment with 12 people, it was found that some faced difficulty to identify the characters “q”, “y” and “0”. This indicates that the reading interface of the Braille needs to be improved.

B. Benefit

Reading braille with the finger tips, studies have shown, stimulates the visual cortex, and so it is that readers of braille feel the same intimate word-to-brain connection as do their sighted peers [3].

At present in Bangladesh the only solution available for a blind person is external help. The goal of this paper is to eliminate this with the help of technology. Around the world complete stop solution for Braille exists from Companies like DOT,

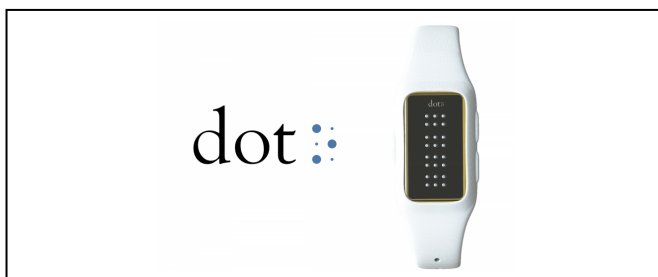


Fig. 6. A DOT Braille watch [8].

The cost of DOT Braille reaches near \$300 which is not feasible for majority of people.

As 90% of the visually impaired people live in developing or underdeveloped country, this cost-effective Braille could help them to a great extent.

C. Future Work

In the future, we plan to make the device real-time device using live feed as an input. To do that, we need to attach a camera with it and use Optical Character Recognition (OCR) techniques. It can be possible using MATLAB's OCR library and syncing it with the Arduino Uno. Furthermore, we plan to include Bengali alphabets recognition in it. Object recognition can also be included if we need to help the blind to move about freely. Using live feed, the device will be more feasible and helpful for the blind students for study purpose and outdoor movement.

VI. CONCLUSION

The device will be a very useful tool to educate the mostly underprivileged blind people to turn them into a valuable workforce. As the device was made using low-cost materials, the poor people of the country would be able to afford it for the purpose of their education. The most expensive part of the device is the Arduino Uno which costs around \$8. So the visually disabled people will be able to overcome the curse of illiteracy. This device is electrically insulated so there is no chance of users getting electric shocks. Moreover, the dots are covered with small plastic rings so that it does not hurt the user's finger tips.

REFERENCES

- [1] (August 2014). visual impairment and blindness. retrieved from <http://www.who.int/mediacentre/factsheets/fs282/en/>
- [2] L. Nahar, A. Jaafar, E. Ahamed & A. B. M. A. Kaish, "Design of a Braille learning application for visually impaired students in Bangladesh" The Official Journal of RESNA, Vol. 27, Issue 3, pp. 172-182, May 2015.
- [3] D. Kendrick. (July 2009). Product evaluation lowering the price of Braille. Retrieved from <http://www.afb.org/afbpress/pubnew.asp?DocID=aw100405>
- [4] Technology Recource List. Retrieved from <https://nfb.org/technology-resource-list#OBWDevices>
- [5] R. Gupta, P. K. Singh and S. Bhanot "Design and Implementation of Arduino based Refreshable Braille Display Controller" Indian Journal of Science and Technology, Vol 9(33), September 2016
- [6] S. Kelly Braille. Technology: What's New and Emerging? Retrieved from <http://www.visionaware.org/info/everyday-living/essential-skills/reading-writing-and-vision-loss/braille-technology-7770/1235>
- [7] Retrieved from <http://www.brailleauthority.org/tg/web-manual/images/appAbraille-ascii.jpg>
- [8] C. Moreshead. (July 2015). This Korean startup is making wearables for the blind, but its ambition is much bigger Retrieved from <https://www.techinasia.com/dot-braille-smartwatch>